

Research Evaluation

Audiometer

Make/Model: GSI-61

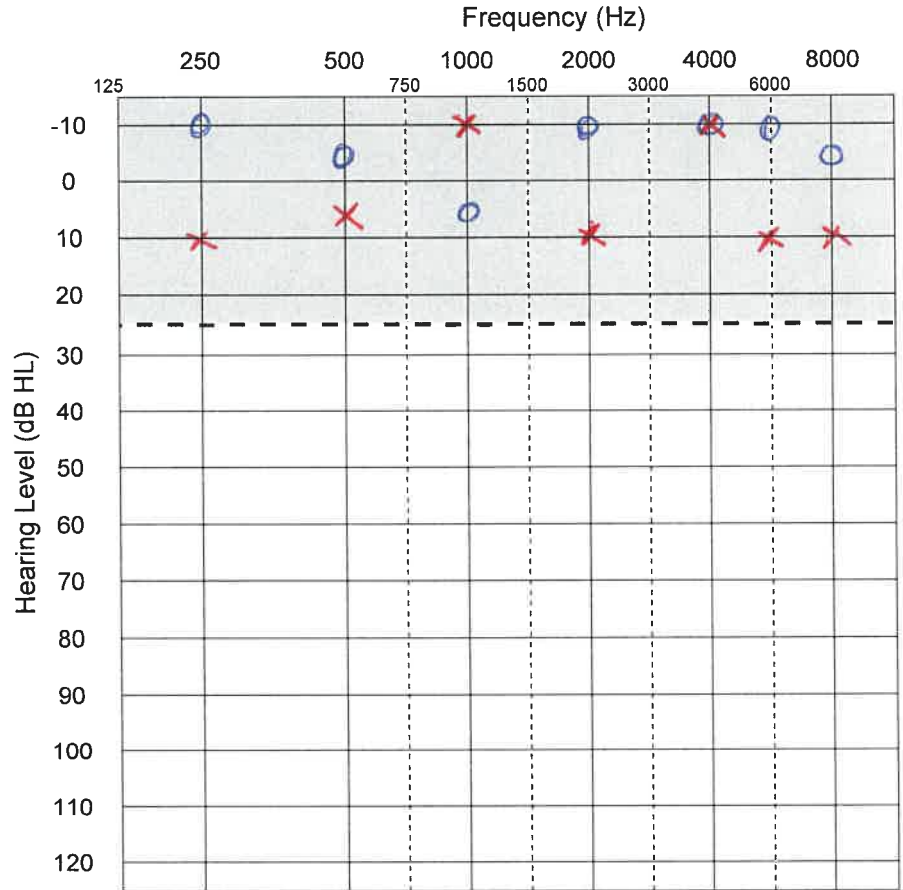
Reliability: Good Fair Poor

PT Stimulus: Pulsed Steady FM

Transducer: Inserts TDH-50 Other: _____

Audiogram Key	Ear		
	Left	Unspecified	Right
Air Conduction (AC)	X		O
Masked AC	□		Δ
Bone Conduction (BC)	>	^	<
Masked BC	]		[
Sound Field AC (Unaided)		S	
Sound Field AC (Aided)		A	
Sound Field AC (Implant)		I	
No Response	↙	↓	↘

Acoustic Immitance					
		Right Ear	Left Ear		
Volume (cc)					
Compliance (cm ³)					
Peak Pressure (daPa)					
Gradient (daPa)					
Acoustic Reflex Threshold					
Probe	Stim	Frequency (kHz)			Decay
		.5	1	2	
Right	IPSI				
	Contra				
Left	IPSI				
	Contra				



Masking Level											
		250	500	750	1000	1500	2000	3000	4000	6000	8000
RE	AC										
	BC										
LE	AC										
	BC										

Speech Audiometry									
	PTA	SRT	Speech Discrimination		MCL	UCL	Lists		
	HL	HL	%	HL	%	HL			
RE									
Masking LE									
LE									
Masking RE									

Remarks:

ID NH273	RESEARCH STUDY Ro1	AGE
SITE OF EXAMINATION OHRC	SIGNATURE OF EXAMINING AUDIOLOGIST	DATE OF EXAM

FOR RESEARCH PURPOSES ONLY. FINDINGS OBTAINED IN ACCORDANCE WITH & DETERMINED BY SPECIFIC STUDY PROTOCOL. NOT INTENDED FOR HEALTH CARE MANAGEMENT.

Word Familiarity Test – Brief A

For each word below, circle the number that best reflects how familiar you are with that word.

1 = Never seen/heard 2 = Might have seen/heard 3 = Pretty sure you've seen/heard

4 = Recognize but don't know meaning 5 = Seen, vague idea of meaning

6 = Think you know meaning 7 = Confident you know meaning

1. inrush 1 (2) 3 4 5 6 7
2. dysprosium 1 2 3 (4) 5 6 7
3. chemurgic (1) 2 3 4 5 6 7
4. cenobitic 1 2 3 4 (5) 6 7
5. obi (1) 2 3 4 5 6 7
6. bosh 1 2 (3) 4 5 6 7
7. batrachian (1) 2 3 4 5 6 7
8. bemire 1 (2) 3 4 5 6 7
9. appanage 1 2 3 (4) 5 6 7
10. citify 1 2 (3) 4 5 6 7
11. puttee (1) 2 3 4 5 6 7
12. arable (1) 2 3 4 5 6 7
13. ferrule 1 (2) 3 4 5 6 7
14. meliorate 1 2 3 (4) 5 6 7
15. capstan (1) 2 3 4 5 6 7
16. hidalgo (1) 2 3 4 5 6 7
17. crosier 1 (2) 3 4 5 6 7
18. ennui 1 2 3 4 5 (6) 7
19. exegesis 1 2 3 4 (5) 6 7
20. mullion 1 2 3 4 (5) 6 7
21. fief 1 2 (3) 4 5 6 7
22. triumphvir (1) 2 3 4 5 6 7
23. mastoid 1 2 3 4 5 (6) 7
24. hemolytic 1 2 3 (4) (5) 6 7
25. scintillate 1 2 3 4 5 (6) 7
26. czarina 1 (2) 3 4 5 6 7
27. warder 1 2 3 4 5 (6) 7
28. darnel (1) 2 3 4 5 6 7
29. grandiloquence 1 2 3 4 5 (6) 7
30. coitus 1 2 3 4 5 6 (7)
31. triennial 1 2 3 4 5 (6) 7
32. smirch 1 2 3 (4) 5 6 7
33. concertina 1 2 3 (4) 5 6 7
34. rasher 1 2 (3) 4 5 6 7
35. ruse 1 2 3 4 5 6 (7)
36. audiophile 1 2 3 4 5 6 (7)

37. perspicuous 1 2 3 (4) 5 6 7
38. stolidly 1 2 3 4 5 (6) 7
39. Pullman 1 2 3 (4) 5 6 7
40. vesture 1 2 3 (4) 5 6 7
41. underslung 1 2 3 4 5 (6) 7
42. refutation 1 2 3 4 5 6 (7)
43. briquette 1 2 3 4 5 6 (7)
44. pommel 1 2 3 (4) 5 6 7
45. knobbed 1 2 3 4 5 6 (7)
46. affray 1 2 3 (4) 5 6 7
47. genic 1 2 3 4 5 (6) 7
48. freedman 1 2 3 4 5 (6) 7
49. autumnal 1 2 3 4 5 6 (7)
50. goodwill 1 2 3 4 5 6 (7)
51. deluge 1 2 3 4 5 6 (7)
52. defy 1 2 3 4 5 6 (7)
53. invigorate 1 2 3 4 5 6 (7)
54. euphemism 1 2 3 4 5 6 (7)
55. dike 1 2 3 4 5 6 (7)
56. forked 1 2 3 4 5 6 (7)
57. drag 1 2 3 4 5 6 (7)
58. handrail 1 2 3 4 5 6 (7)
59. gab 1 2 3 4 5 6 (7)
60. destitute 1 2 3 4 5 6 (7)
61. antler 1 2 3 4 5 6 (7)
62. cannibal 1 2 3 4 5 6 (7)
63. quit 1 2 3 4 5 6 (7)
64. objectivity 1 2 3 4 5 6 (7)
65. comforter 1 2 3 4 5 6 (7)
66. greed 1 2 3 4 5 6 (7)
67. grab 1 2 3 4 5 6 (7)
68. mother 1 2 3 4 5 6 (7)
69. central 1 2 3 4 5 6 (7)
70. glory 1 2 3 4 5 6 (7)
71. battery 1 2 3 4 5 6 (7)
72. leaves 1 2 3 4 5 6 (7)

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CVC-Normal Hearing (NH) Adult Data Sheet

1/7

Task	Date completed	Initials
Consent Form (circle): Aim 2b		
HIPPA		
Mini Mental State Exam (MMSE) score: _____		
Audiogram (unaided PT AC & BC, SRT & WRS), Otoscopy & Tympanometry		
Vocabulary Chart	8/14/25	W/S

Questionnaire:

Date: 8/19/25

Hearing History

1. Which is your dominant hand? ☐ RIGHT ☐ LEFT ☒ AMBIDEXTROUS

2. Do you have perfect pitch? ☐ YES ☒ NO

01 / 2025
Month Year

3. When was the last time you had your hearing tested?

4. Do you have problems hearing? ☐ YES ☒ NO ☐ UNSURE

4a. In which ear? ☐ RIGHT only ☐ LEFT only ☐ BOTH

4b. When did you first notice problems with your hearing? _____

5. Do you have any family members with hearing loss? ☐ YES ☒ NO

5a. If yes, list family members, ages, and etiologies: _____

6. Do you have a history of noise exposure? ☐ YES ☒ NO

6a. If yes, please describe type of noise exposure, length of exposure/day/week/month/year, and number of years of exposure: _____

Subject ID: NH273

Age: _____

7. Do you experience tinnitus (sounds within your ear)? ☐ YES ☒ NO ☐ UNSURE

7a. In which ear? ☐ RIGHT only ☐ LEFT only ☐ BOTH

Month / Year

7b. When did you first notice the tinnitus?

7c. How would you describe the sound of your tinnitus? _____

7e. Are you currently experiencing tinnitus? ☐ YES ☐ NO ☐ UNSURE

7d. How often does this occur?

☐ ALWAYS

☐ DAILY ☐ times/day

☐ WEEKLY ☐

times/week

☐ MONTHLY ☐

times/month

8. Have you previously participated in hearing-related research experiments? ☒ YES ☐ NO

8a. If yes, describe the experiment and number of hours? Vocal experiment

Music & Language Background

9. Are you a Native English Speaker? ☒ YES ☐ NO

9a. If yes; other spoken languages: _____

9b. If yes; speaks American English: ☒ YES ☐ NO

9c. If yes; area you grew up in: Indiana, Indianapolis

10. How many hours per day or week do you listen to music through headphones? 0 hrs/ day or week

11. Do you have a musical background (such as taken singing, voice, and/or instrumental music lessons, sing in a choir or play a musical instrument)? ☒ YES ☒ NO

Subject ID: NH273

Age: _____

11a. Length of your musical background: 1 years11b. Describe your musical background: ~~Choir~~ Choir**Audiogram**

1) Copy down thresholds from audiogram into table below

Frequency	250	500	1000	1500	2000	3000	4000	6000
Left Θ								
Right Θ								

2) Insert audiogram as new text document into subjects folders in the C:drive and save as Audiogram.txt

- Date Audiogram.txt document made: _____

Vowel Introduction

☐ Go over the laminated vowel word sheet and introduce vowels (IH,EH, and AE). Explain how these are synthetic vowels and may sound a little different then use to (subject may keep laminated sheet in site during testing)

Learning Tasks

Date: _____

1) **Vocabulary Size Test**

Trial	Total Score (L):
1	<u>361</u>

Subject ID: NH273

Age: _____

2) Binaural Same Vowel CVC

MATLAB command: CVCmono('subjectID', 'B', 66, 'y');

	.txt filename	% correct (B):
1	Experiments/Data/ NH273/NH273 -CVCmono- 0.txt	84.85
2	Experiments/Data/ NH273/NH273 -CVCmono- 1.txt	90.91
3	Experiments/Data/ NH273/NH273 -CVCmono- 2.txt	89.39

3) Monaural Same Vowel CVC left ear → Subject has 3 chances to score $\geq 85\%$; Must do at least 2 trials; If there is a learning effect ($>5\%$ improvement), then repeat once at the end.

MATLAB command: CVCmono('subjectID', 'L', 66, 'y');

1	Experiments/Data/ NH273/NH273-CVCmono- 3.txt	92.42
2	Experiments/Data/ NH273/NH273-CVCmono- 4.txt	90.91
3	Experiments/Data/	
4	Experiments/Data/	

4) Monaural Same Vowel CVC right ear → Subject has 3 chances to score $\geq 85\%$; Must do at least 2 trials

MATLAB command: CVCmono('subjectID', 'R', 66, 'y');

Trial	.txt filename:	% correct (R):
1	Experiments/Data/ NH273/NH273-CVCmono- 5.txt	93.94
2	Experiments/Data/ NH273/NH273-CVCmono- 6.txt	95.45
3	Experiments/Data/	
4	Experiments/Data/	

* Note that the fourth row is used for an additional optional repeat

Ask the subject if they have any comments or if they heard any words/vowels that were not an option (After first and last trial)...

Notes: Sounds less robotic; git sounds like gir
 always heard full words; that C2 sounds off smother
 g.g → gld; deaf vs das
 git/get similar lead better than led

Monaural Qualitative Questions (Ask after general questioning at the end of the right ear trial):

- Clarity of words: pretty clear; ~~that~~ C2 sometimes mumbled
- Vowels/Sounds heard: IH, EH, AE (maybe one extra sound)
- Clarity of Each Type of Vowel: 1. IH: IH 2. EH: EH AE 3. AE: AE EH
- How many vowels heard each time: 1

Subject ID: NH273

Age: _____

Experimental Tasks (5 runs)

Date: _____

1) **CVC (dichotic)** MATLAB command: CVC('subjectID',66,0,44);

Trial	.txt filename:	% both correct:	% >=1 correct:
1	/Experiment/Data/ <u>NH273/NH273_CVC-0.txt</u> <i>Remember to ask about participant's experience and note it in the General Notes section afterwards</i>	<u>63.64</u> % lowlow <u>77.27</u> % highhigh <u>68.18</u> % alt Average: <u>69.70</u> %	<u>90.91</u> % lowlow <u>100</u> % highhigh <u>100</u> % alt Average: <u>96.97</u> %
2	/Experiment/Data/ <u>NH273/NH273_CVC-1.txt</u>	<u>68.18</u> % lowlow <u>86.36</u> % highhigh <u>77.27</u> % alt Average: <u>77.27</u> %	<u>95.45</u> % lowlow <u>100</u> % highhigh <u>100</u> % alt Average: <u>98.48</u> %
3	/Experiment/Data/ <u>NH273/NH273_CVC-2.txt</u> <i>Remember to provide a 5-10 minute break</i>	<u>86.36</u> % lowlow <u>81.82</u> % highhigh <u>86.36</u> % alt Average: <u>84.54</u> %	<u>95.45</u> % lowlow <u>100</u> % highhigh <u>100</u> % alt Average: <u>98.48</u> %
4	/Experiment/Data/ <u>NH273/NH273_CVC-3.txt</u>	<u>72.73</u> % lowlow <u>81.82</u> % highhigh <u>81.82</u> % alt Average: <u>78.74</u> %	<u>100</u> % lowlow <u>100</u> % highhigh <u>100</u> % alt Average: <u>100</u> %
5	/Experiment/Data/ <u>NH273/NH273_CVC-4.txt</u>	<u>50.00</u> % lowlow <u>86.36</u> % highhigh <u>68.18</u> % alt Average: <u>68.18</u> %	<u>100</u> % lowlow <u>100</u> % highhigh <u>100</u> % alt Average: <u>100</u> %
—	/Experiment/Data/ This is an optional placeholder for redos.	_____ % lowlow _____ % highhigh _____ % alt Average: _____ %	_____ % lowlow _____ % highhigh _____ % alt Average: _____ %

Subject ID: NH273

Age: _____

Ask the subject if they have any comments. **Confirm how they annotate and use the IH, EH and AE vowel buttons.** Did they hear any sounds/vowels that were not an option? Did the sound/vowel ever lateralize (ex: only heard something from the right headphone and nothing from the left) (Ask after first and fifth trial)?...

General Notes: Not that many IH ; thee sounds: word blends
sounds more clear

lidge + AE - close to ledge

lth + EH close to leth

did + AE

CONTINUE ON BACK →

Subject ID: NH273

Age: _____

Dichotic Qualitative Questions (Ask after general questioning at the end of the fifth trial):

1. Clarity of words/ Consonants: Couldnt tell ending sound existed
2. Vowels/Sounds heard (provide examples):
IH, AE, EH
3. Fusion types:
 - a. Heard Three Sounds (provide examples): Maybe once or twice
 - b. Heard Full word + Start of another word (provide examples): Yes, beginning sounds in left ear
 - c. Heard Full word + Only Vowel (provide examples): Yes,
 - d. Heard Differing End Consonants/ Sounds (provide examples): Yes ; dad
4. When would you choose the word+vowel below it?: heard echoes, word + vowel
5. Lateralization (Were the different sounds always heard on different sides of the ears or was one of them heard in the center and the other heard on one side):
No (has difficulty with localization)
6. Did any side ever feel louder?: Don't know
7. Did anything make it tricky?: Lots of AE sounds, hard to separate AE & EH; separating git & get
8. Did hearing something like a real word pull your answer one way?: did not matter
9. What was your approach to answering these questions?: 1. try to distinguish between 2 vs. one sound 2. pick word 2. repeat & select vowel
3. check

☐ All tasks data backed up: Date: _____ Turn attn. back to 3 and turn off equipment. Date: _____

Subject ID: NH273

Age: _____

